

How Local Averaging Moves Matérn Pairwise Range Targets: A Smoothness-Dependent Phase Law

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Abstract

Spatial measurements often represent local averages, but are subsequently fitted as if they were point observations. This changes the population target of covariance estimation, not merely its finite-sample bias. We study a stationary Matérn Gaussian field in $\mathbb{R}^d$, observed through a known compact symmetric averaging kernel of bandwidth h . For a fixed nonzero-lag Gaussian two-point likelihood with fixed known smoothness ν , we characterize the exact pseudo-variance and pseudo-decay and derive their small-support expansions. The naive decay shift is of order $h^{2\nu}$ for $0 < \nu < 1$, $h^2 \log(1/h)$ at $\nu = 1$, and h^2 for $\nu > 1$. Explicit coefficients show that, for every $\nu > 0$ and any nondegenerate compact symmetric kernel, naive point-support fitting underestimates inverse range—and therefore overestimates spatial range—for all sufficiently small h . The coefficient for $\nu > 1$ remains positive for anisotropic averaging kernels. A support-aware covariance propagates both the field and measurement noise through the observation operator and removes the population shift when the operator and fixed nuisance parameters are correctly specified. Deterministic two-dimensional quadrature verifies the three rate regimes, while synthetic finite-grid likelihood experiments illustrate discrete, irregular, and boundary behavior. The result isolates a concrete inferential cost of ignored spatial support while making no novelty claim for the standard change-of-support covariance formula itself.

1 Introduction

Spatial values recorded on pixels, grid cells, administrative regions, or after kernel preprocessing are local aggregates of an underlying field. The change-of-support literature models such observations by integrating the latent process over their supports [3, 6]. In practice, however, smoothed values are often assigned to declared centers and a point-support covariance is fitted. Relative to the latent parameters, this is a support-misinterpreted model. A single two-point covariance can still be matched exactly by different variance and decay parameters; a multi-lag likelihood is genuinely misspecified and selects a Kullback–Leibler (KL) projection [2].

That qualitative conclusion is familiar, but it does not answer three basic questions. How fast does the target move as the support shrinks? Does the rate depend on Matérn smoothness? Is the direction of the range error stable across dimension and averaging kernels? These questions matter because the Matérn covariance changes regularity at the origin as ν crosses one. A symmetric kernel cancels first-order Taylor terms at nonzero lags, but it cannot cancel a cusp or fractional power at the origin.

Temporal aggregation of autoregressive and Ornstein–Uhlenbeck models is classical [1, 11], and integrated Ornstein–Uhlenbeck inference has been studied directly [4]. Consequently, a one-dimensional exponential calculation alone is not a novelty claim. Reviews and geostatistical methods

for incompatible supports establish the block-to-block covariance and area-to-point prediction problems [7, 8]. Most closely, Fuentes [5] constructs a likelihood for block averages in terms of an underlying point covariance and studies a Matérn example. That work correctly models the support; it does not derive the pseudo-range selected when block averages are instead treated as points. Generic misspecified-GP KL theory [2] supplies the projection framework but not the support-specific coefficient or sign. Our contribution is the explicit all-smoothness, d -dimensional fixed-lag pseudo-parameter expansion. In particular:

1. We derive a phase law for the naive Matérn decay target: orders $h^{2\nu}$, $h^2 \log(1/h)$, and h^2 in the rough, threshold, and smooth regimes.
2. We give explicit coefficients. Origin expansions, together with a Bessel-function identity in the smooth regime, prove that normalized correlation increases, hence inverse range decreases, for every $\nu > 0$ and any compact symmetric kernel when h is sufficiently small.
3. We distinguish the continuous interior theorem from discrete, row-normalized, and boundary-truncated smoothing. We then compute finite-grid full-likelihood KL targets and compare naive and support-aware fits using only synthetic data.

The covariance transformation $SKS^\top$, block kriging, generic KL projection under misspecification, and temporal aggregation are prior work; they are not claimed as contributions. Recent finite-grid Matérn likelihood work also treats discretization and edge effects without studying the target induced by a preprocessing smoother [9]. Our narrower focus is the smoothness-dependent target displacement caused by ignoring observation support.

2 Model and pseudo-target

2.1 Latent field and local support

Let $Y = \{Y(t) : t \in \mathbb{R}^d\}$ be a mean-zero stationary Gaussian field with

$$\text{Cov}\{Y(t), Y(t+r)\} = C(r) = v\mathcal{M}_\nu(\alpha\|r\|), \quad v > 0, \quad \alpha > 0, \quad \nu > 0, \quad (1)$$

where

$$\mathcal{M}_\nu(x) = \frac{2^{1-\nu}}{\Gamma(\nu)} x^\nu K_\nu(x), \quad \mathcal{M}_\nu(0) = 1. \quad (2)$$

Thus α is an inverse range; larger α means shorter dependence. This convention must not be mixed with the common length-scale parameterization $\alpha = \sqrt{2\nu}/\ell$.

Let k be a symmetric probability density supported on $B(0, L) \subset \mathbb{R}^d$, let $k_h(u) = h^{-d}k(u/h)$, and define the mean-square convolution

$$Z_h(t) = \int_{\mathbb{R}^d} k_h(u)Y(t-u) du. \quad (3)$$

If $U, V \stackrel{\text{iid}}{\sim} k$ and $D = U - V$, stationarity and Fubini's theorem give the exact smoothed covariance

$$C_h(r) = v\mathbb{E}[\mathcal{M}_\nu\{\alpha\|r + hD\|\}]. \quad (4)$$

Write

$$\Sigma_k = \mathbb{E}(UU^\top), \quad T_k = \text{tr}(\Sigma_k), \quad m_q = \mathbb{E}\|D\|^q, \quad (5)$$

so that $m_2 = 2T_k$. We assume $T_k > 0$.

2.2 Naive Gaussian pair fit

Fix $r = Re$, with $R > 0$ and $\|e\| = 1$. The smoothed pair has variance $C_h(0)$ and correlation

$$\rho_h(r) = \frac{C_h(r)}{C_h(0)}. \quad (6)$$

A naive point-support model with the correct known ν but unknown variance and decay uses covariance

$$v^\dagger \begin{pmatrix} 1 & \mathcal{M}_\nu(\alpha^\dagger R) \\ \mathcal{M}_\nu(\alpha^\dagger R) & 1 \end{pmatrix}. \quad (7)$$

The true two-by-two covariance belongs to this family. Therefore its Gaussian KL minimizer is exactly

$$v_h^\dagger = C_h(0), \quad \mathcal{M}_\nu(\alpha_h^\dagger R) = \rho_h(r). \quad (8)$$

Existence follows from $0 < \rho_h(r) < 1$ for $r \neq 0$. Positivity follows from $k \geq 0$ and $\mathcal{M}_\nu > 0$. Strict inequality follows because the smoothed field has spectral density $|\widehat{k}(h\omega)|^2 f_\nu(\omega)$, which is positive on a neighborhood of the origin, while $e^{i\omega^\top r}$ is not identically one there. Uniqueness follows from

$$\mathcal{M}'_\nu(x) = -\frac{2^{1-\nu}}{\Gamma(\nu)} x^\nu K_{\nu-1}(x) < 0, \quad x > 0. \quad (9)$$

For completeness, consider a fixed lattice, a lag r belonging to that lattice, and increasing rectangular Følner domains. If the parameter space is compact, the target is unique and interior, correlations are bounded away from one, and the pair log likelihood is continuous with an integrable quadratic envelope, the ergodic theorem gives the uniform law needed for pair-composite consistency at $(v_h^\dagger, \alpha_h^\dagger)$. Overlapping pairs are dependent, so uncertainty requires a Godambe or long-run covariance, not an independence Hessian. We do not use fixed-domain asymptotics to claim separate consistency of variance and range; such a claim would conflict with Matérn microergodicity results [10, 12].

3 Small-support phase law

For $z = \alpha R$, define the directional kernel variance $a_k(e) = e^\top \Sigma_k e$ and

$$B_{\nu,k}(r) = \alpha^2 \left[a_k(e) \mathcal{M}''_\nu(z) + \{T_k - a_k(e)\} \frac{\mathcal{M}'_\nu(z)}{z} \right]. \quad (10)$$

Proposition 1 (Fixed nonzero lag). *For r in a compact annulus bounded away from zero and $h \leq \|r\|/(4L)$,*

$$\frac{C_h(r)}{v} = \mathcal{M}_\nu(\alpha\|r\|) + h^2 B_{\nu,k}(r) + O(h^4). \quad (11)$$

The remainder is uniform when α also ranges over a compact subset of $(0, \infty)$.

The variance behaves differently because the Matérn function is expanded at the origin. Put

$$c_\nu = -\frac{\Gamma(-\nu)}{2^{2\nu}\Gamma(\nu)} = \frac{\Gamma(1-\nu)}{\nu 2^{2\nu}\Gamma(\nu)} > 0, \quad 0 < \nu < 1. \quad (12)$$

Theorem 1 (Matérn support phase law). *Fix $\nu > 0$. Under the preceding assumptions, as $h \downarrow 0$,*

$$\frac{C_h(0)}{v} = 1 - c_\nu m_{2\nu}(\alpha h)^{2\nu} + O(h^2), \quad 0 < \nu < 1, \quad (13)$$

$$\frac{C_h(0)}{v} = 1 - \frac{m_2}{2}(\alpha h)^2 \log\{1/(\alpha h)\} + O(h^2), \quad \nu = 1, \quad (14)$$

$$\frac{C_h(0)}{v} = 1 - \frac{m_2}{4(\nu-1)}(\alpha h)^2 + o(h^2), \quad \nu > 1. \quad (15)$$

Moreover, the naive decay target in (8) satisfies

$$\alpha_h^\dagger - \alpha = \frac{\mathcal{M}_\nu(z) c_\nu m_{2\nu} \alpha^{2\nu}}{R \mathcal{M}'_\nu(z)} h^{2\nu} + O\left(h^{\min(4\nu, 2)}\right), \quad 0 < \nu < 1, \quad (16)$$

$$\alpha_h^\dagger - \alpha \sim \frac{m_2 \mathcal{M}_1(z) \alpha^2}{2R \mathcal{M}'_1(z)} h^2 \log(1/h), \quad \nu = 1, \quad (17)$$

$$\alpha_h^\dagger - \alpha = \frac{A_{\nu,k}(r)}{R \mathcal{M}'_\nu(z)} h^2 + \mathcal{R}_\nu(h), \quad \nu > 1, \quad (18)$$

where

$$A_{\nu,k}(r) = \alpha^2 \{(2\nu-2)a_k(e) + T_k\} G_\nu(z) > 0, \quad (19)$$

$$G_\nu(z) = \frac{\mathcal{M}_\nu(z)}{2\nu-2} + \frac{\mathcal{M}'_\nu(z)}{z} = \frac{2^{1-\nu}}{\Gamma(\nu)} \frac{z^\nu K_{\nu-2}(z)}{2\nu-2} > 0, \quad (20)$$

and $\mathcal{R}_\nu(h) = O(h^{2\nu})$ for $1 < \nu < 2$, $O\{h^4 \log(1/h)\}$ for $\nu = 2$, and $O(h^4)$ for $\nu > 2$. For fixed ν , these remainders are uniform for α in a compact subset of $(0, \infty)$ and r in a compact annulus bounded away from zero. They are not uniform as ν approaches one or two.

Every leading coefficient in (16)–(18) is negative because $\mathcal{M}'_\nu(z) < 0$. Thus, for sufficiently small positive h ,

$$v_h^\dagger < v, \quad \rho_h(r) > \mathcal{M}_\nu(\alpha R), \quad \alpha_h^\dagger < \alpha. \quad (21)$$

The naive model overestimates range. The result does not require an isotropic smoother: anisotropy enters only through $a_k(e)$ and T_k , whose combination in (20) remains positive. For an isotropic kernel with $\Sigma_k = \sigma_k^2 I_d$,

$$A_{\nu,k}(r) = \sigma_k^2 \alpha^2 (2\nu + d - 2) G_\nu(z). \quad (22)$$

Remark 1 (Why symmetry does not imply an h^2 bias). Symmetry removes the order- h term at a fixed nonzero lag. When $0 < \nu < 1$, however, the origin expansion contains $\|r\|^{2\nu}$, which changes the variance at order $h^{2\nu}$. Normalizing by that variance dominates the order- h^2 off-diagonal change. At $\nu = 1$, the origin term is $h^2 \log(1/h)$. Only for $\nu > 1$ do ordinary second derivatives control both pieces.

Remark 2 (The fitted lag is fixed). The expansion holds for R bounded away from zero. If $R = ch$, the nonzero-lag Taylor expansion is no longer uniform and the relative target displacement need not vanish. No shrinking-lag conclusion is claimed.

4 Support-aware finite-grid likelihood

The continuous theorem concerns translation-invariant interior averaging. The finite experiment starts from raw sites $X_{\text{in}} = (x_1, \dots, x_n)$ and a rectangular row-stochastic operator $S_h \in \mathbb{R}^{m \times n}$. If independent measurement noise is present before smoothing,

$$y = S_h\{Y(X_{\text{in}}) + \epsilon\}, \quad \epsilon \sim N(0, \tau^2 I_n), \quad (23)$$

then the correct covariance is

$$\Sigma_{\text{corr}}(\alpha) = S_h\{K_\alpha(X_{\text{in}}, X_{\text{in}}) + \tau^2 I_n\}S_h^\top. \quad (24)$$

The naive declared-center point-support model instead uses

$$\Sigma_{\text{naive}}(\alpha) = K_\alpha(X_{\text{out}}, X_{\text{out}}) + \tau^2 I_m. \quad (25)$$

Besides ignoring support, (25) incorrectly treats pre-smoothing noise as independent post-smoothing noise. With variance, smoothness, and nugget fixed at their generating values, the finite-design population target minimizes

$$Q_n(\alpha) = \frac{1}{2} [\log \det \Sigma(\alpha) + \text{tr}\{\Sigma(\alpha)^{-1} \Sigma_0\}], \quad (26)$$

where constants independent of α are omitted. The corrected family contains Σ_0 exactly, provided the known operator and fixed nuisance parameters are correct and α is identifiable on the finite design; the naive family generally does not. Our primary support-only experiment sets $\tau = 0$, preventing noise timing from confounding the support effect. A separate combined-misspecification run with $\tau^2 = 0.05$ is retained as a reproducibility stress artifact but is not used for the principal claims.

A rectangular S_h is intentional. If a square parameter-independent smoother is invertible, the exact transformed likelihood is merely a change of coordinates and need not lose information. Rectangular aggregation represents the scientifically relevant reduction from raw locations to fewer supported measurements.

Boundary row normalization is not covered by the interior theorem. For two location-dependent scaled kernels with means μ_s and μ_t , a direct fixed-lag expansion gives

$$C_{h,s,t}(r) = C(r) + h \nabla C(r)^\top (\mu_s - \mu_t) + O(h^2). \quad (27)$$

A truncated kernel can therefore introduce an order- h term and destroy stationarity. The interior theorem provides no universal boundary sign. We report one boundary-retaining and one fixed jittered design as illustrations, not as a boundary scaling study.

5 Synthetic validation

5.1 Continuous two-dimensional quadrature

We used a product Epanechnikov density

$$k(u) = \prod_{j=1}^d \frac{3}{4} (1 - u_j^2) \mathbf{1}\{|u_j| \leq 1\}. \quad (28)$$

Two-dimensional Matérn support-bias phase law

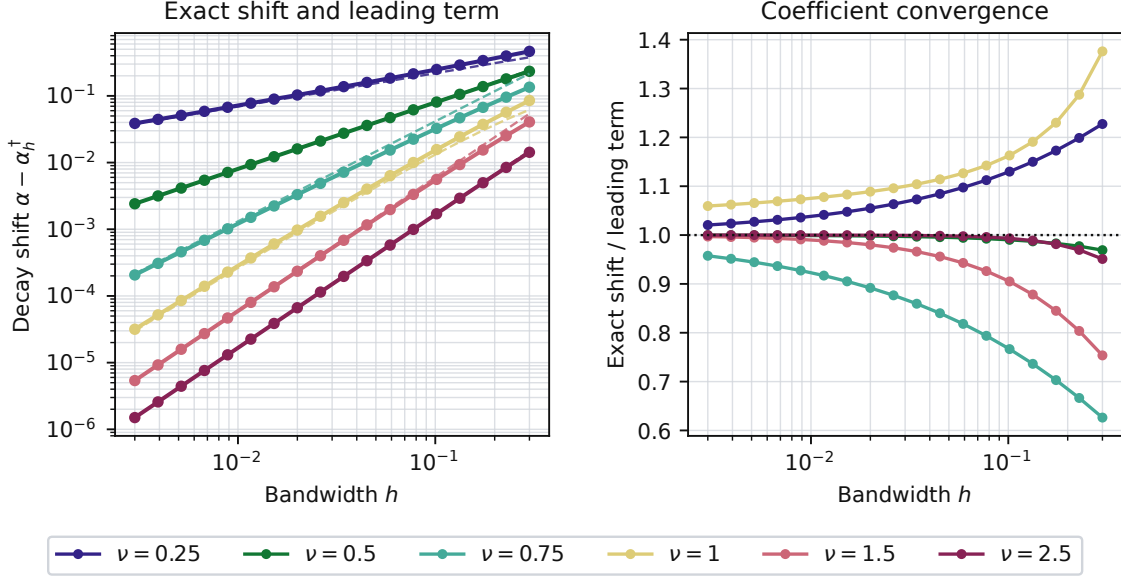


Figure 1: Continuous two-dimensional pseudo-decay shift under product Epanechnikov smoothing. Left: exact shifts (solid) and theorem leading terms with their explicit coefficients (dashed). Right: their ratio, which approaches one as $h \downarrow 0$.

The one-dimensional density of $U_j - V_j$ is available in closed form, so tensor Gauss–Legendre quadrature evaluates (4) deterministically. We used $d = 2$, $\alpha = R = 1$, quadrature order 96 per coordinate, 18 geometrically spaced bandwidths from 0.003 to 0.3, and $\nu \in \{0.25, 0.5, 0.75, 1, 1.5, 2.5\}$. For each cell we solved (8) by monotone root finding. Repeating the entire grid at quadrature orders 64 and 128 changed pseudo-decay by at most 4.4×10^{-8} and 5.2×10^{-9} , respectively, relative to order 96.

All 108 computed decay shifts $\alpha - \alpha_h^\dagger$ were positive. An ordinary least-squares slope over the six smallest bandwidths on log–log axes was 0.515, 0.999, and 1.468 for $\nu = 0.25, 0.5, 0.75$, compared with theoretical powers 0.5, 1, and 1.5. At $\nu = 1$, the raw slope was 1.817, as expected from the logarithmic modifier rather than a pure power. The slopes for $\nu = 1.5$ and 2.5 were 1.994 and 2.000. Figure 1 displays the full paths and the coefficient-level theorem comparison. At the smallest bandwidth, exact-shift to leading-term ratios were 1.021, 1.000, 0.958, 1.059, 0.997, and 1.000 in increasing order of ν .

5.2 Finite-grid full likelihood

The core design uses a 19×19 input lattice with spacing 0.25 and every second location as a candidate output center. Interior outputs are trimmed by 0.75 in each coordinate. We set $v = \alpha = 1$, $\tau^2 = 0$, $\nu \in \{0.5, 1, 1.5, 2.5\}$, and $h \in \{0, 0.3, 0.5, 0.7\}$. Radial Epanechnikov rows are normalized exactly. The zero-bandwidth operator is an exact rectangular selector. Stress panels retain boundary centers, perturb each input coordinate uniformly in $[-0.04, 0.04]$ before clipping to the original bounding box, and examine one-dimensional increasing-domain lattices with 100, 225, and 400 raw sites. The irregular panel uses one fixed jitter realization.

For each configuration, the finite-design population criterion is minimized by bounded scalar

Table 1: Finite-design validation at $h = 0.7$. The table is generated from immutable task shards; no failed or boundary-hitting fit is silently removed.

ν	Model	Target	Mean	MCSE	Bound hits
0.5	Corrected	1.000	1.090	0.047	3
0.5	Naive	0.152	0.151	0.004	0
1	Corrected	1.000	0.990	0.010	0
1	Naive	0.414	0.407	0.005	0
1.5	Corrected	1.000	0.990	0.007	0
1.5	Naive	0.573	0.570	0.005	0
2.5	Corrected	1.000	0.994	0.004	0
2.5	Naive	0.740	0.738	0.003	0

optimization over $\log \alpha \in [\log(0.02), \log(5)]$. Replicate likelihoods are evaluated on a 161-point equally spaced log-decay profile; an interior grid minimum is refined by three-point quadratic interpolation, while an endpoint minimum is retained as a boundary fit. Gaussian raw-observation vectors with the declared covariance are transformed by S_h . Configurations sharing all parameters except h reuse standard-normal draws. No numerical jitter is added. A Cholesky or population-optimization failure aborts its configuration and makes the reducer mark the run incomplete; no such failure occurred.

The promoted study used 200 replicates in each of 21 configurations, producing 8,400 fitted estimates. The reducer verified configuration hashes, row counts, and unique model–replicate keys for every task shard. Independent checks also verified finite numerical fields, expected model labels, replicate indices, and metadata. Across the 16 core two-dimensional configurations, the corrected population target differed from the true decay by at most 1.3×10^{-7} . At the largest bandwidth, $h = 0.7$, the naive population targets were 0.152, 0.414, 0.573, and 0.740 for $\nu = 0.5, 1, 1.5, 2.5$, respectively. Their corresponding Monte Carlo means were 0.151, 0.407, 0.570, and 0.738; the corrected means were 1.090, 0.990, 0.990, and 0.994. The comparatively noisy corrected estimate at $\nu = 0.5$ nevertheless remained within two Monte Carlo standard errors of its population target.

No optimizer or covariance failure was discarded. Six of the 6,400 core fits landed at the declared bounds, and those fits remain in Figure 2 and Table 1. Including boundary outputs changed both the output dimension and the rough-field naive target, from 0.152 to 0.135 at $h = 0.7$; for the fixed jittered design the target changed from 0.289 to 0.284 at $h = 0.5$. In both stress cases the corrected target remained one. In the one-dimensional domain-growth check, increasing the number of raw sites from 100 to 400 reduced the corrected estimator standard deviation from 0.529 to 0.177 and the naive standard deviation from 0.151 to 0.068, while at $n = 400$ their means remained within 0.026 and 0.009, respectively, of their population targets (Figure 3).

6 Discussion

The theorem separates two effects that are often conflated. Linear propagation of covariance through a known support is standard. The new object here is the parameter selected when that propagation is ignored. For rough Matérn fields, the origin singularity makes the pseudo-target move at a fractional power of bandwidth. For smooth fields, a Bessel recurrence shows that the combined variance and nonzero-lag correction still increases normalized correlation, including under anisotropic smoothing.

Several limitations define the scope of the result. First, ν and the support are treated as known. Jointly estimating smoothness, bandwidth, variance, and range may be weakly identified. Second,

the closed target is for a two-point composite fit. A full likelihood cannot generally match every smoothed lag; its KL target is computed numerically in our finite designs. Third, fixed-resolution discrete averaging does not converge to continuous convolution merely because the spatial domain grows. A mixed-domain quadrature limit or an exact discrete-weight theorem is required. Fourth, boundary normalization creates location-dependent supports and can introduce first-order terms. Fifth, the synthetic study fixes nuisance parameters so that it tests the central target-shift mechanism rather than a larger optimization problem.

These restrictions are preferable to an unsupported broad claim. They also suggest direct extensions: a uniform discrete-to-continuous approximation, joint nuisance profiling, spectral full-likelihood expansions, and explicit boundary counterexamples. Without such extensions, the present contribution is best viewed as a focused methodological paper rather than a high-dimensional statistics result.

7 Conclusion

For a fixed nonzero-lag pair fit under translation-invariant interior averaging, ignored support makes a Matérn field look more persistent than it is. The magnitude is controlled by smoothness: $h^{2\nu}$, $h^2 \log(1/h)$, or h^2 . A support-aware observation covariance removes the finite-design population shift when the operator is known, nuisance parameters are correctly specified, and decay is identifiable. The result provides a theorem-level diagnostic for when preprocessing is negligible and when it changes the pairwise range estimand; it makes no universal claim for boundary supports or arbitrary full-likelihood fits.

A Proof of the phase theorem

Proof. Let $f_\alpha(x) = \mathcal{M}_\nu(\alpha\|x\|)$. Since r is bounded away from zero and D is bounded, a fourth-order Taylor expansion is uniform. Because $D = U - V$ has a centrally symmetric law, both $\mathbb{E}D = 0$ and $\mathbb{E}(D^{\otimes 3}) = 0$; moreover $\mathbb{E}(DD^\top) = 2\Sigma_k$. Therefore

$$\mathbb{E}f_\alpha(r + hD) = f_\alpha(r) + \frac{h^2}{2} \text{tr}\{\nabla^2 f_\alpha(r) \mathbb{E}(DD^\top)\} + O(h^4). \quad (29)$$

For $r = Re$,

$$\nabla^2 f_\alpha(r) = \alpha^2 \mathcal{M}_\nu''(z) ee^\top + \alpha^2 \frac{\mathcal{M}_\nu'(z)}{z} (I - ee^\top), \quad (30)$$

which proves (11).

For noninteger ν , the origin expansion is

$$\mathcal{M}_\nu(x) = \sum_{j=0}^{\infty} \frac{(x^2/4)^j}{j!(1-\nu)_j} + \frac{\Gamma(-\nu)}{2^{2\nu}\Gamma(\nu)} x^{2\nu} \sum_{j=0}^{\infty} \frac{(x^2/4)^j}{j!(1+\nu)_j}. \quad (31)$$

Taking expectations at $x = \alpha h\|D\|$ yields (13) and (15). At $\nu = 1$,

$$\mathcal{M}_1(x) = 1 + \frac{x^2}{2} \{\log(x/2) + \gamma_E - 1/2\} + O\{x^4(1 + |\log x|)\}, \quad (32)$$

which proves (14).

At the second transition,

$$\mathcal{M}_2(x) = 1 - \frac{x^2}{4} + x^4 \left\{ \frac{3}{64} - \frac{\log(x/2) + \gamma_E}{16} \right\} + O\{x^6(1 + |\log x|)\}. \quad (33)$$

This gives an $O\{h^4 \log(1/h)\}$ variance remainder at $\nu = 2$. For fixed $\nu > 2$, four-times differentiability at the origin gives

$$\mathcal{M}_\nu(x) = 1 - \frac{x^2}{4(\nu-1)} + O(x^4).$$

For $1 < \nu < 2$, the fractional term in (31) is $O(x^{2\nu})$.

Divide the fixed-lag expansion by the relevant variance expansion and apply a first-order Taylor expansion to the inverse of $\alpha \mapsto \mathcal{M}_\nu(\alpha R)$. This directly gives (16) and (17); the same division and inverse expansion preserve the stated $O(h^{2\nu})$, $O\{h^4 \log(1/h)\}$, and $O(h^4)$ remainders in the three smooth subregimes. For $\nu > 1$, the order- h^2 coefficient of normalized correlation is

$$B_{\nu,k}(r) + \mathcal{M}_\nu(z) \frac{\alpha^2 m_2}{4(\nu-1)}. \quad (34)$$

The Matérn correlation satisfies

$$\mathcal{M}_\nu''(z) + \frac{1-2\nu}{z} \mathcal{M}_\nu'(z) - \mathcal{M}_\nu(z) = 0. \quad (35)$$

Using $m_2 = 2T_k$, the coefficient becomes $A_{\nu,k}(r)$. Finally,

$$K_\nu(z) = K_{\nu-2}(z) + \frac{2(\nu-1)}{z} K_{\nu-1}(z) \quad (36)$$

implies (20). Positivity of $K_\eta(z)$ for real η and $z > 0$, together with (9), proves the sign. $\square$

B Exact exponential benchmark

For $d = 1$, $\nu = 1/2$, and the Epanechnikov density $k(u) = \frac{3}{4}(1 - u^2)\mathbf{1}\{|u| \leq 1\}$, the density of $D = U - V$ is

$$g(w) = \frac{3}{5} - \frac{3}{4}|w|^2 + \frac{3}{8}|w|^3 - \frac{3}{160}|w|^5, \quad |w| \leq 2. \quad (37)$$

With $x = \alpha h$, define

$$s(x) = \mathbb{E}e^{-x|D|}, \quad M_k(x) = \mathbb{E}e^{xU} = \frac{3(x \cosh x - \sinh x)}{x^3}, \quad q(x) = M_k(x)^2. \quad (38)$$

Then $C_h(0) = vs(x)$, and for $R \geq 2h$,

$$C_h(R) = vq(x)e^{-\alpha R}. \quad (39)$$

The exact naive pair target is

$$v_h^\dagger = vs(x), \quad \alpha_h^\dagger = \alpha - \frac{1}{R} \log \frac{q(x)}{s(x)}. \quad (40)$$

For $x > 0$, compact support and nondegeneracy give

$$1 < q(x)/s(x) < e^{2x}.$$

Indeed, condition on $A = |U|$ and $B = |V|$, and put $p = A+B$ and $m = |A-B|$. Symmetry of the signs gives conditional numerator and denominator proportional to $\cosh(xp) + \cosh(xm)$ and $e^{-xp} + e^{-xm}$, respectively. The lower inequality follows from $\cosh(t) \geq e^{-t}$, with strictness off $A = B = 0$. Since $p + m = 2 \max(A, B) \leq 2$, $\cosh(xp) < e^{xp} \leq e^{2x-xm}$ and $\cosh(xm) \leq e^{xm} \leq e^{2x-xp}$, with strictness on a set of positive probability. Averaging proves the upper inequality. Since $R \geq 2h$, this implies $0 < \alpha - R^{-1} \log\{q(x)/s(x)\} < \alpha$. Variance and decay are therefore both strictly underestimated. The small-bandwidth expansions are

$$s(x) = 1 - \frac{18}{35}x + \frac{1}{5}x^2 - \frac{4}{63}x^3 + \frac{3}{175}x^4 + O(x^5), \quad (41)$$

$$q(x) = 1 + \frac{1}{5}x^2 + \frac{3}{175}x^4 + O(x^6), \quad (42)$$

$$\log\{q(x)/s(x)\} = \frac{18}{35}x + \frac{162}{1225}x^2 + O(x^3). \quad (43)$$

This benchmark is used as an exact software regression test, not as the paper’s novelty claim.

C Reproducibility contract

The public code at <https://github.com/Sasha-Cui/HighDimSpatialStatistics> is archived for this draft under the immutable tag `paper-draft-v0.1.0`. It uses an immutable JSON manifest, SHA-256 configuration hashes, deterministic seed derivation, one task-local atomic shard, and an `afterany` reducer that refuses to mark a run complete when a shard is missing or invalid. The zero-bandwidth selector, row sums, row rank, corrected population target, grid interpolation, and closed exponential formulas are unit tested. The reported implementation adds no adaptive diagonal jitter. All paper figures and tables are regenerated from audited aggregate CSV files.

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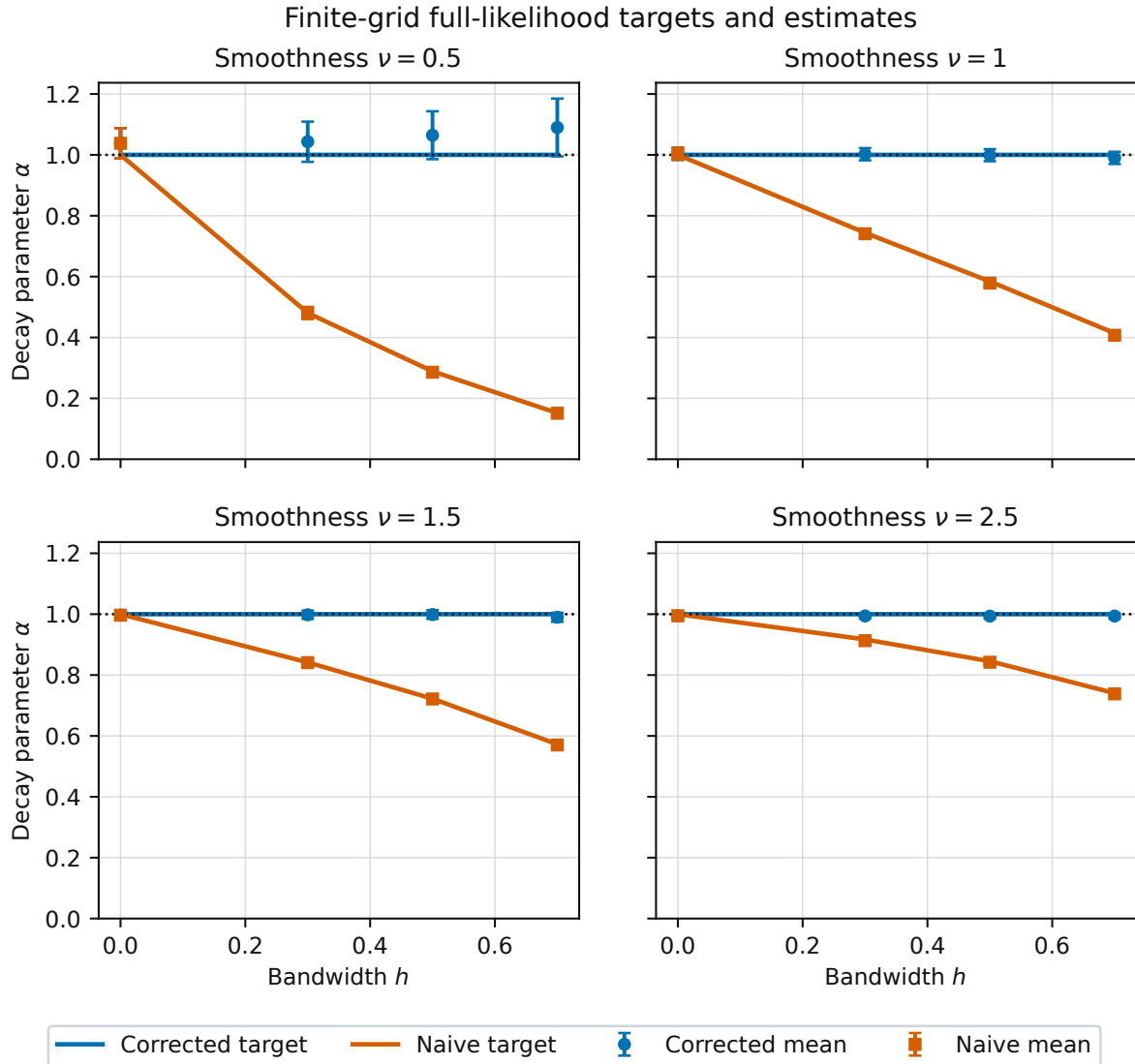


Figure 2: Finite-design population targets and Monte Carlo mean decay estimates for corrected and naive likelihoods. Error bars show two Monte Carlo standard errors of the mean, not estimator confidence intervals.

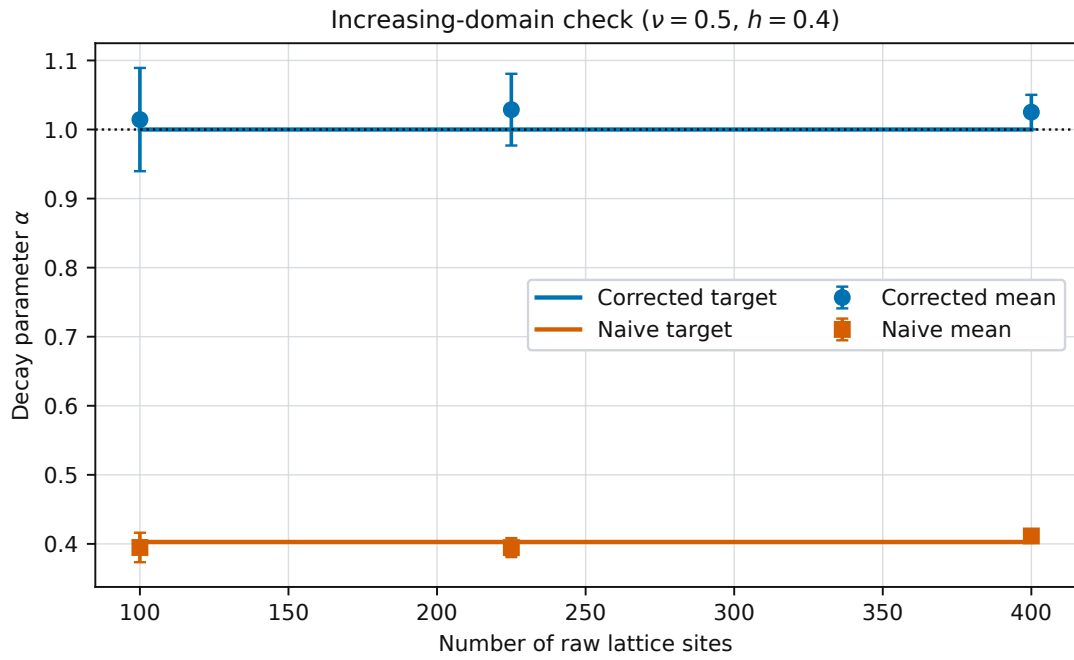


Figure 3: Increasing-domain check for the finite-grid likelihood. Lines are nonrandom finite-design targets and points are Monte Carlo means from 200 replicates; error bars show two Monte Carlo standard errors of each mean.